Remember, Buffett does not use an equity risk premium in his calculations. Instead, he adjusts for riskiness by the margin of safety represented by the price he is willing to pay.

Using this theory, we can make our own calculation of IBM's value. Using a two-stage dividend discount model, I assumed IBM would grow its cash earnings at 7 percent for 10 years and 5 percent after that. I then discounted these cash flows by 10 percent—substantially higher than the 2 percent of the 10-year U.S. Treasury note. The higher discount rate simply builds in a greater margin of safety. Based on these calculations, IBM is worth $326 per share, much higher than the average $169 price Buffett paid. If we adjust the growth rate for the next 10 years down to 5 percent, closer to Gartner's estimate of the IT services industry's growth, the value is $279 per share, still $100 per share higher than what Buffett paid.

Another way to look at the valuation question is to ask what growth rates are embedded in a $169 share price. To be worth $169 per share, IBM would have to grow its owner earnings at 2 percent in perpetuity. Readers might quibble at a $326 or $279 estimate of fair value for a share of IBM. But just as many, I suspect, would argue that IBM will grow faster than 2 percent per year over the next decade. The fair value answer, I am sure, lies somewhere in between these two estimates, reminding us again of one of Buffett's favorite maxims: "I would rather be approximately right than precisely wrong."

In many ways, the IBM purchase reminds me of Buffett's purchase of the Coca-Cola Company. At the time, many critics were puzzled. Buffett had bought the stock near its all-time high (just like IBM). Many believed Coca-Cola was a boring, slow-growing company whose best days were behind it (just like IBM). When Buffett purchased Coca-Cola, the company's share price was 15 times earnings and 12 times cash flow, a 30 percent and 50 percent premium to the market average. When we ran the dividend discount model on Coca-Cola's owner earnings using various growth rates, we found the company was selling for a significant discount to fair value despite the premium price-to-earnings and price-to-cash-flow ratios. Assuming an unbelievably low 5 percent growth rate for Coca-Cola, the dividend discount model said Coca-Cola was worth $20.7 billion—far higher than its current market value of $15.1 billion.